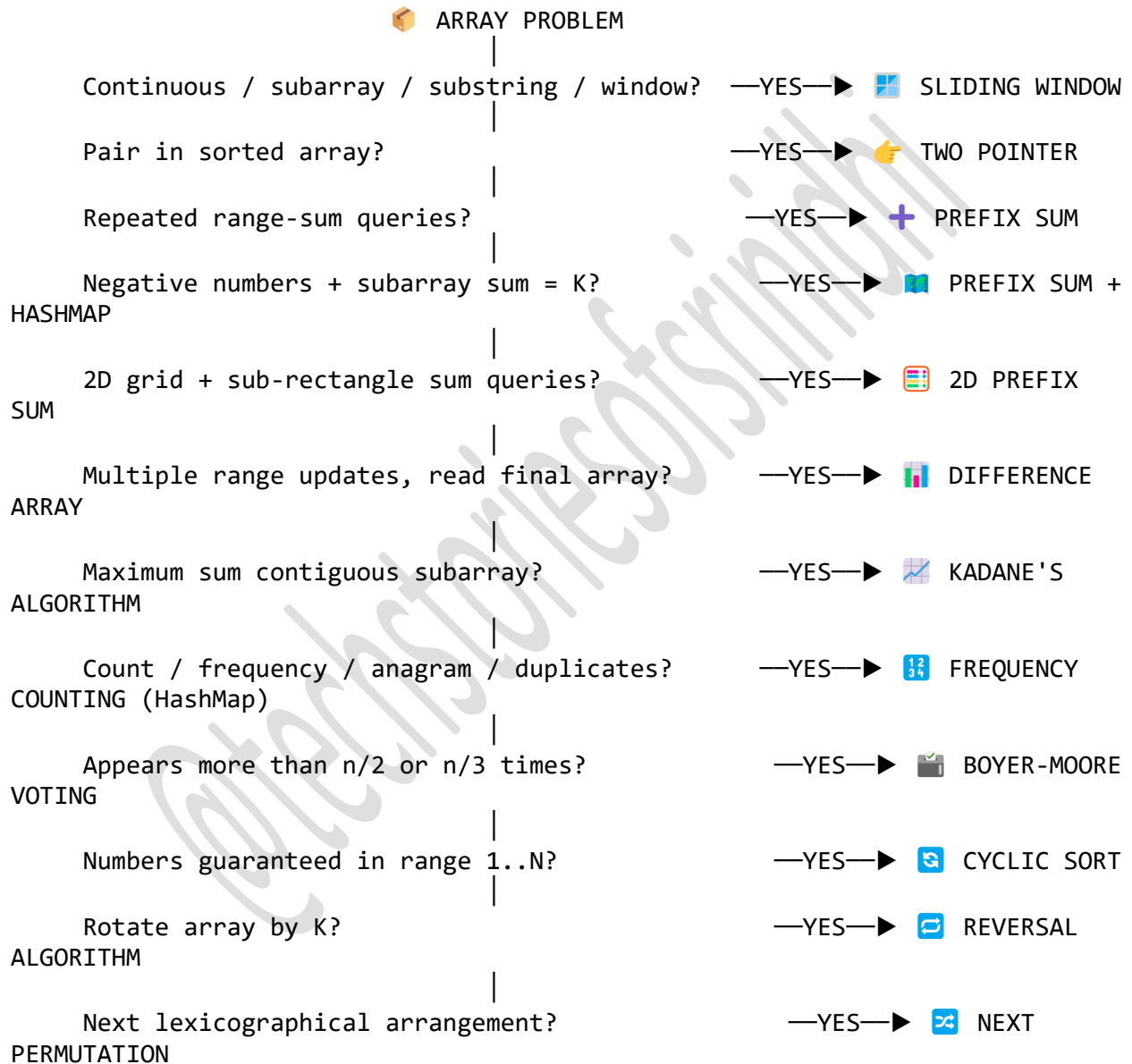


⚡ Array Interview Cheatsheet

Quick Revision — Print & Read Before Every Interview

🌳 Pattern Decision Tree (Read Top to Bottom)



Trigger Words → Pattern Lookup

Trigger Words in the Question	Pattern
"contiguous", "consecutive", "subarray", "substring", "window of size K"	Sliding Window
"sorted array", "pair", "palindrome", "container", "merge two sorted"	Two Pointer
"sum from index i to j", "multiple queries"	Prefix Sum
"subarray sum = K", "contains negatives"	Prefix Sum + HashMap
"matrix", "sub-rectangle sum"	2D Prefix Sum
"range update", "multiple bookings"	Difference Array
"maximum sum subarray"	Kadane's Algorithm
"most/least frequent", "anagram", "duplicate"	Frequency Counting
"more than n/2 times", "more than n/3 times"	Boyer-Moore Voting
"numbers from 1 to N", "missing number", "duplicate number"	Cyclic Sort
"rotate by K"	Reversal Algorithm
"next permutation", "next arrangement"	Next Permutation

Time & Space Complexity Table

Pattern	Time	Space
Sliding Window	$O(n)$	$O(1) / O(k)$
Two Pointer	$O(n) / O(n^2)$ triplets	$O(1)$
Prefix Sum	$O(n)$ build, $O(1)$ query	$O(n)$
Prefix Sum + HashMap	$O(n)$	$O(n)$
2D Prefix Sum	$O(\text{rows} \times \text{cols})$	$O(\text{rows} \times \text{cols})$
Difference Array	$O(1)$ update, $O(n)$ rebuild	$O(n)$
Kadane's Algorithm	$O(n)$	$O(1)$
Frequency Counting	$O(n)$	$O(k)$
Boyer-Moore Voting	$O(n)$	$O(1)$
Cyclic Sort	$O(n)$	$O(1)$
Rotate Array	$O(n)$	$O(1)$
Next Permutation	$O(n)$	$O(1)$

One-Line Mental Models

Pattern	Say This to Yourself
Sliding Window	"Grow right, shrink left when condition breaks."
Two Pointer	"Move pointers toward each other based on sum comparison."
Prefix Sum	"Precompute running totals so range queries are $O(1)$."
Prefix Sum + HashMap	"Has <code>currentSum - K</code> occurred before?"
2D Prefix Sum	"Inclusion-exclusion on rectangles."
Difference Array	"Mark start (+V) and end+1 (-V), reconstruct later."
Kadane's Algorithm	"Extend or restart — whichever gives a bigger sum."
Frequency Counting	"Count first, then answer instantly."
Boyer-Moore Voting	"Cancel out non-majority votes; majority always survives."
Cyclic Sort	"Put every number at its correct index."
Rotate Array	"Reverse all, then reverse the two parts."
Next Permutation	"Find pivot, swap with next-larger, reverse suffix."

Top 10 Mistakes to Avoid (Quick List)

1. Not clarifying constraints before coding (sorted? negatives? duplicates?)
2. Confusing subarray (contiguous) with subsequence (not contiguous)
3. Forgetting `{0: 1}` base case in Prefix Sum + HashMap
4. Forgetting to shrink the window when a condition breaks
5. Not normalizing `k = k % n` before rotating
6. Assuming a majority element always exists
7. Using $O(n)$ space (HashMap) when $O(1)$ space (Cyclic Sort / Boyer-Moore) was expected
8. Skipping the brute-force explanation before jumping to optimal code
9. Not testing code with an example before saying "I'm done"
10. Not stating time/space complexity out loud after coding

Final Checklist Before Interview

- ☐ I can name all 12 patterns from memory
- ☐ I can match trigger words to patterns instantly
- ☐ I know the Big-O for each pattern
- ☐ I can code each pattern's core template without notes
- ☐ I always explain brute force → optimal out loud
- ☐ I always test my code with an example

Companion to "The Complete Array Interview Handbook" — keep this cheatsheet handy for last-minute revision.